COS80022 – Software Quality and Testing

Library Book Reservation System - Requirements Analysis Document

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Introduction

1.1 Purpose of the System

The purpose of the Library Book Reservation System is to streamline the process of locating, reserving, and managing books for both library members and librarians. It eliminates inefficiencies of manual reservations, reduces errors, and improves user satisfaction by providing an accessible and transparent reservation service. The system ensures fairness through automated waitlist management and timely notifications.

1.2 Scope of the System

The system will serve three main groups of stakeholders:
• Library Members: Search for books, place reservations, track status, and receive alerts.
• Librarians: Manage reservations, oversee waitlists, override bookings if necessary, and maintain records.
• System Administrators: Ensure performance, security, and integration with other services.
The scope includes features such as online search, digital reservation requests, automated notification handling, waitlist management, and role-based access for staff. It excludes payment handling and digital content distribution.

1.3 Objectives and Success Criteria

• Reduce manual workload of librarians by at least 60%.
• Ensure book availability and reservation status are always accurate in real time.
• Allow users to reserve and check status within 3 clicks or under 1 minute.
• Successfully handle 500+ concurrent users during peak times.

1.4 Definitions

Reservation: A system-generated request that holds a book for a specific member.
Waitlist: A prioritized queue of users requesting an unavailable book.
Notification: A system-generated alert sent via email or SMS.
Pickup Window: The maximum period (e.g., 3 days) for which a reserved book is held.

1.5 References

- Software Quality Standards- https://en.wikipedia.org/wiki/ISO/IEC_9126
- Software Requirements Specification- https://ieeexplore.ieee.org/document/720574
- City of Melbourne Libraries – https://www.melbourne.vic.gov.au/borrowing-and-renewing

1.6 Overview

The Library Book Reservation System provides a comprehensive solution for managing reservations. By combining user authentication, automated reservation handling, and librarian oversight, the system enhances the efficiency of book circulation, improving accessibility and user experience.

2. Current System

Currently, reservations are managed manually. Users must call or visit the library to reserve books. This process often leads to delays, human errors, lack of transparency, and uninformed users about reservation changes. It is inefficient, time-consuming, and unsuitable for modern expectations.

3. Proposed System

3.1 Overview

The proposed system is a web- and mobile-based platform that automates reservations. Users log in with their credentials, search for books, and place reservations in minimal steps. The system validates credentials, checks availability, and confirms a reservation or places the user in a waitlist. Notifications are automatic, and librarians have a dashboard for management tasks.

3.2 Functional Requirements

• Book Search: Search by title, author, genre, ISBN, keywords, and availability.
• User Authentication: Login with card number and PIN; recovery via email.
• Reservation Handling: Reserve books marked 'Available'.
• Waitlist Handling: Queue users for unavailable books on a first-come basis.
• Notifications: Alerts by email and SMS for availability, expiry, or cancellation.
• Reservation Expiry: Hold reserved books for 3 business days.
• Librarian Privileges: Override reservations, manage waitlists, generate reports.
• Multi-copy Support: Manage reservations across multiple copies.

3.3 Non-functional Requirements

3.3.1 Usability

- The system should be easy for both users and librarians to use.
- The design should be simple and consistent across web and mobile.
- It should work for people with accessibility needs .
- Error messages should be clear and helpful.

3.3.2 Reliability

- The system should be available whenever the library is open.
- Reservation and user data should be backed up every day.
- If the system goes down, it should be restored quickly.
- Data should not get lost or duplicated.

3.3.3 Performance

- Searching for books should be fast.
- The system should handle many users at the same time.
- Notifications should be sent quickly after a reservation or change.

3.3.4 Supportability

- The system should work on common browsers.
- Library staff should be able to change simple settings without needing technical staff.
- A simple help guide should be available for new users.

3.3.5 Implementation

- The system should be built in a way that allows adding new features later.
- It should use a reliable database for storing books and reservations.
- It should connect with the library's existing catalog.

3.3.6 Interface

- The system should be able to send notifications by email and SMS.
- It should allow secure login for users.
- It may support external login systems (like university login) if needed in future.

3.3.7 Packaging

- The system will be available as a web application.
- The installation package should include setup instructions.

3.3.8 Legal

- It should respect library rules for borrowing and reservations.
- User data should be protected and not shared without permission.

3.4 System Models

3.4.1 Scenarios

Scenario 1: Reserving a Book (Library Member)

- The user logs into the library system.
- They search for a book by title, author, or subject.
- The system shows availability.
- If available, the user clicks **Reserve**.
- The system confirms the reservation and sends a notification (email/SMS).
- The librarian prepares the book for collection.

Scenario 2: Joining a Waitlist (Library Member)

- The user searches for a book that is currently fully reserved.
- The system offers an option to **Join Waitlist**.
- The user confirms.
- The system places them in the waitlist queue.
- When the book is returned, the system notifies the next user in line.

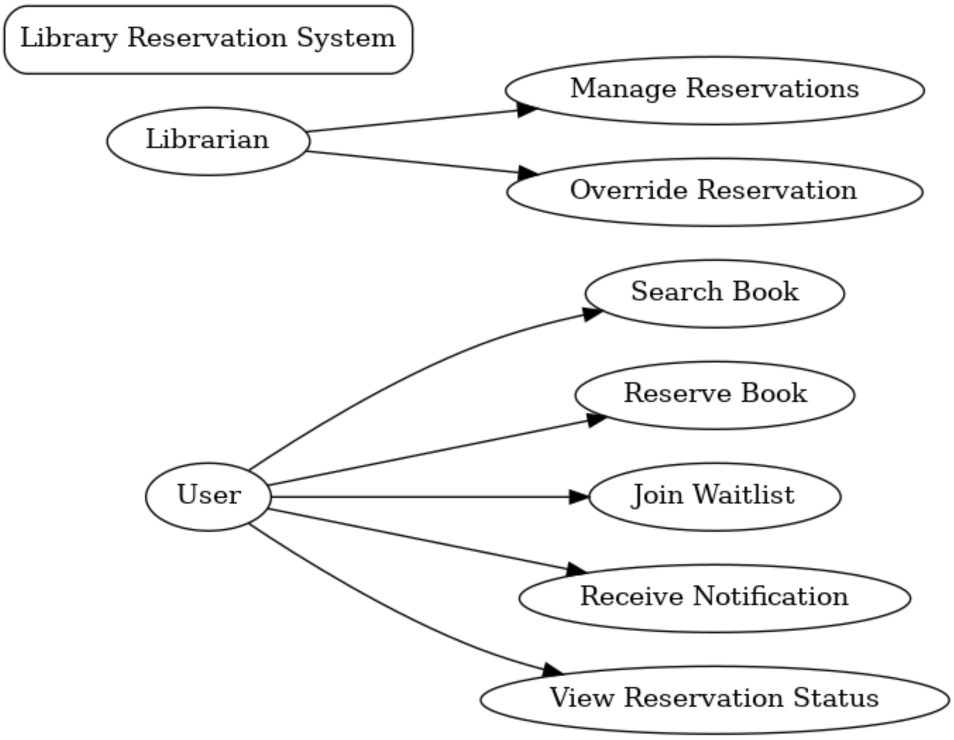
Scenario 3: Cancelling a Reservation (Library Member)

- The user logs into their account.
- They view their current reservations.
- They select a reservation and click **Cancel**.
- The system frees up the copy.
- If there is a waitlist, the system automatically notifies the next person.

Scenario 4: Managing Reservations (Librarian)

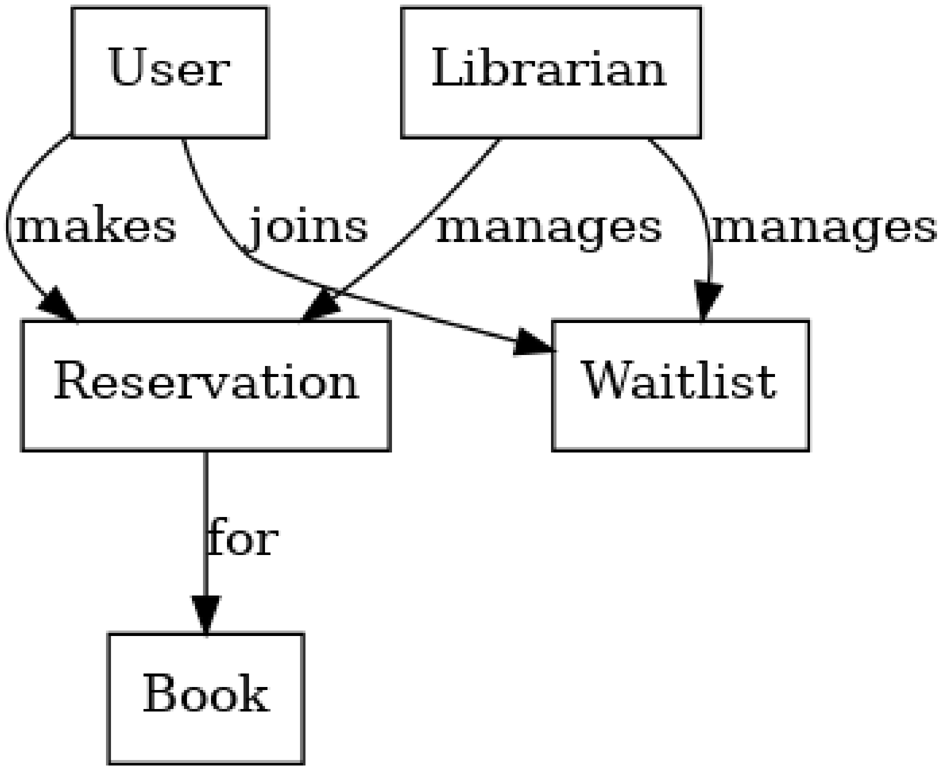
- The librarian logs into the system with admin access.
- They check the list of reservations and waitlists.
- The librarian marks reserved books as **Ready for Pickup** once available.
- They also cancel expired reservations if users don't collect books on time.
- Reports on reservations and waitlists can be generated for management.

3.4.2 Use Case Diagram



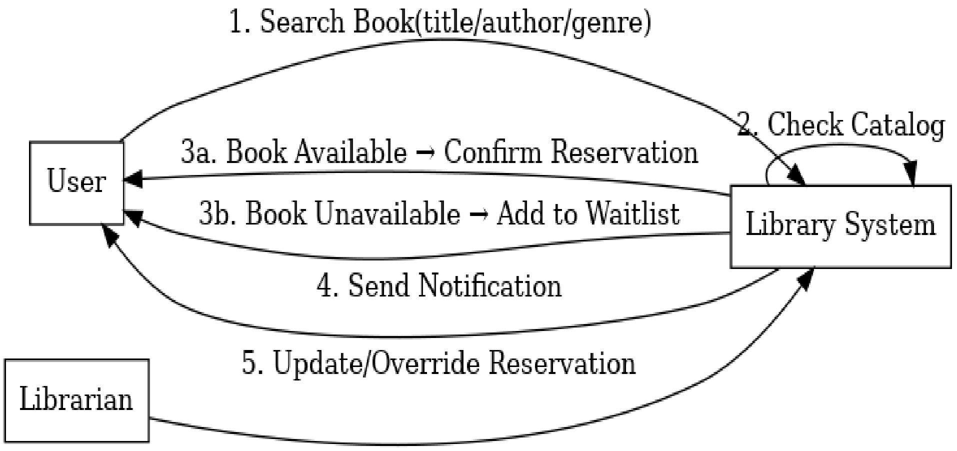
This diagram shows how Users and Librarians interact with the system. Users can search, reserve, join waitlists, receive notifications, and view reservations, while Librarians manage and override reservations.

3.4.3 Object Model



This simplified class diagram shows the relationships: Users make Reservations for Books, join Waitlists when unavailable, and Librarians manage both Reservations and Waitlists.

3.4.4 Dynamic Model



This sequence diagram illustrates the process: the User searches for a book, the System checks availability, then either reserves the book and notifies the User, or places them in the Waitlist. Librarians can update or override reservations.

3.4.5 User Interface

The system will have a simple and intuitive interface designed for two main types of users: library members and librarians.

For Library Members:

- **Login Page:** Users log in using their member ID and PIN.
- **Search Page:** Users can search books by title, author, or subject.
- **Book Details Page:** Shows availability, allows users to reserve or join a waitlist.
- **My Account Page:** Displays active reservations, waitlist status, and history.
- **Notifications Page:** Shows alerts .

For Librarians:

- **Admin Login Page:** Secure login with librarian credentials.
- **Dashboard:** Overview of reservations, waitlists, and overdue reservations.
- **Manage Reservations Page:** Allows librarians to mark books as ready, cancel expired holds, or view reports.
- **User Management Page:** Allows librarians to update member records.

4. Glossary

PIN: Personal Identification Number for authentication.
Reservation: Request to hold a book copy.
Waitlist: Queue for unavailable books.
Notification: Automated message to users.
Pickup Window: Time limit for reserved books.